

Liu Runzhi

Last Updated on 23rd July 2025

liu.r.020@ms.saitama-u.ac.jp | 090-8497-5918

Education

Saitama University

MS in Computer Science
Saitama, Saitama, Japan
2025.04-2027.03 (Expected)

Nihon University

BE in Computer Science
GPA: 3.27 / 4.0
kouriyama,fukusima, Japan
2021.04-2025.03

Links

Github:// **Asocialist**

Coursework

Graduate

Computer Network
Computer Vision
Cryptography
Data Mining
Software Engineering
Information Theory
English Speaking

Undergraduate

Data Structure and Algorithms
Databases
Python Programming
Computer Graphics

Skills

Programming

Over 3 years
C • C++ • Python
Over 1 year
Java • Javascript • HTML • $\LaTeX$
Less than 1 year
Markdown • SQL • Shell

Technologies

Familiar
Git • Linux • Ros • Ros2 • Vue.js

Software

Familiar
Github • IntelliJ IDEA •
Microsoft Office

Profile

Master's student in Computer Science at Saitama University, focusing on the integration of Large Language Models with mobile robotics. Experienced in building intelligent systems involving path planning, perception, and human-robot interaction. Skilled in Python, C++, ROS/ROS2, and deep learning frameworks.

Passionate about scalable, customer-focused software development. Eager to contribute to Amazon's innovation-driven environment by applying strong problem-solving abilities, a hands-on engineering mindset, and alignment with Amazon's Leadership Principles.

Experience

Nihon University Bachelor of Engineering in Computer Science

2021.4 - 2025.03

- Studied a wide range of topics including Computer Vision, Deep Learning, Point Cloud Processing, Data Handling, Robotics Development, Web Design, Embedded Systems, Data Structures, Information Theory, Automata Theory, and Human-Computer Interaction.
- Acquired hands-on experience in algorithm design and implementation for real-world robotic applications.

Saitama University Master of Engineering, Graduate School of Science and Engineering

2025.04-2027.03

- Conducting research on the integration of Large Language Models (LLMs) with mobile robotics, exploring ways to enhance robot decision-making and human-robot interaction using advanced AI models.
- Investigating how LLMs can support high-level task planning, natural language understanding, and adaptive behavior in autonomous systems.

Researchs

Adaptive Path Planning for Autonomous Mobile Robots

2024-2025

- Developed a path planning system for autonomous mobile robots that dynamically switches algorithms based on obstacle density in the environment.
- Implemented multiple pathfinding algorithms (A^* , Θ^*) and designed a decision mechanism to select the most efficient one in real-time.
- Gained hands-on experience in robot simulation and algorithm tuning using tools such as ROS and Python/C++.

Enhancing Human-Robot Interaction with Large Language Models

2025-

- Conducting research on improving natural language interaction between humans and mobile robots using ChatGPT API.
- Developing an autonomous shopping assistant robot capable of understanding human instructions and navigating environments to fulfill shopping-related tasks.
- Integrated ChatGPT API with robotic systems to enable dynamic task interpretation, dialogue-based clarification, and adaptive behavior.

Awards

2025	Scholarship	The Takazawa Sanjiro International Scholarship
2024	3th Prize	Academic Scholarship in Nihon University
2023	3th Prize	Academic Scholarship in Nihon University

Language

English	Fluent	TOEIC: 770pt
Japanese	Fluent	JLPT: N1
Chinese	Native	